Public Sentiment Institute — Complete Reference

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Attribute	Detail
Pathway name	Pathway 3 — Dual Universe
Best use case	High-scrutiny publications, peer review, competitive races requiring full methodological independence between RV and LV universes
Min sample n	1,000+ respondents (Set B: 1,200+)
Weighting set	Set B preferred (41 cells, Age×Education joint dimension)
Effort level	High — approximately 2× the computation of Pathway 1; separate target maintenance required
Defensibility	Highest — fully independent universes, Monte Carlo CI, LV shift decomposition
Authored by	Public Sentiment Institute · Wisconsin + PSI + SOCAL frameworks

Document scope: This reference covers the complete Pathway 3 methodology from first principles — beginning with how pre-survey benchmarks are established before a single respondent is contacted, through the Likely Voter propensity scoring questions and their mathematical treatment, through dual-universe target derivation, through the full weighting pipeline, and finally through the application architecture that implements every step for end users.

PART I

Complete Pathway 3 Dual Universe Methodology

The methodology is presented in the exact order of execution. Every design decision is explained alongside the step that depends on it. Three analytical frameworks contribute distinct components: Wisconsin provides the raking engine, diagnostics, and recall calibration; PSI provides the Likely Voter propensity model; and SOCAL provides the Bayesian target derivation procedure.

What makes Pathway 3 uniquely defensible: In most polling pipelines, the Likely Voter universe inherits the Registered Voter weighting — the LV screen is applied after RV weighting is complete. Pathway 3 is architecturally different: after a shared data foundation step, the pipeline splits into two completely independent tracks. Each universe gets its own benchmark targets, its own entropy balancing initialization, its own iterative raking pass, and its own recall calibration. No computation from the RV track is reused in the LV track. The result is two universes that can each be published and defended independently.

PIPELINE ARCHITECTURE — EXECUTION ORDER

Phase	What happens	Universe	Framework
0. Pre-survey benchmarks	Establish RV and LV demographic targets BEFORE fieldwork opens. Lock as fixed decisions.	Both	SOCAL
1. Data foundation	Ingest CSV, quality screen, build demographic variables, compute LV scores on unweighted raw sample, assign Q2 history buckets.	Shared	Wisconsin + PSI
2. Post-field target update	SOCAL credibility-weighted update of pre-survey targets using observed sample composition.	Both (separate)	SOCAL
3a. RV track	Cell collapse → Entropy balancing → 4-round raking → two-stage recall calibration → propensity scoring → weight_rv.	RV only	Wisconsin
3b. LV track	Cell collapse → Entropy balancing (P(vote) seeded) → 4-round raking → voter-only recall → weight_lv.	LV only	Wisconsin + PSI
4. Monte Carlo	3×3 scenario grid (3 turnout levels × 3 LV target sets = 9 runs). Reports full uncertainty envelope.	LV	PSI
5. Diagnostics + output	DEFF, effective N, covariate balance, bootstrap SE, shift decomposition table, RV and LV tabbooks, external validation.	Both	Wisconsin

Phase 0 — Establishing Pre-Survey Benchmarks

This phase occurs before a single respondent is contacted. It is the most consequential design decision in the methodology because the targets established here anchor every subsequent weight. The SOCAL framework requires that targets be formed from external data sources only — no survey data is used at this stage. Once the field opens, these prior targets are locked.

> **Why pre-survey benchmarks matter:** A pollster who sets demographic targets after seeing the data can unconsciously choose targets that produce a desired result. Locking targets before fieldwork begins is a transparency commitment: the methodology cannot be steered toward a particular outcome. This is the core discipline of the SOCAL framework.

RV Prior Benchmark Formation

The Registered Voter benchmark targets are formed by blending two independent external sources at equal weight. The first source is the ACS/CPS adult population estimate — the Census Bureau's best measurement of who lives in the United States and their demographic characteristics. The second source is the most recent relevant election cycle exit poll (the 2022 midterm exit poll for a 2026 survey, the 2024 presidential exit poll for questions about the presidential electorate). Each source is weighted 50%.

This blend is deliberate: the ACS alone represents all adults, not just voters, and therefore overstates the share of young and non-college respondents in the voting electorate. The exit poll alone reflects who actually voted in a prior cycle, which may not match the current cycle. The 50/50 blend anchors the RV targets to real-world voter participation patterns without over-constraining them to any single prior election.

LV Prior Benchmark Formation

The Likely Voter benchmark targets are formed separately, anchored to the expected electorate for the current cycle rather than the full adult population. The prior blends two prior midterm exit polls at equal weight (2022 at 50%, 2018 at 50%). This two-cycle blend reduces the influence of idiosyncratic wave effects from a single election while capturing genuine structural changes in the midterm electorate over time.

Source Benchmarks Used (2026 Midterm Universe)

Preset	Description	Age 18–29	WNC	WC
PSI Composite (RV)	50% ACS adult population + 50% 2022 midterm exit. Default Pathway 3 RV prior.	~17%	~36%	~29%
2022 Midterm Exit	CNN/NEP actual 2022 midterm electorate. Older, less college than presidential years.	12%	38%	33%
2024 Presidential Exit	CNN/NEP 2024 presidential electorate. Non-college majority (WNC 38% > WC 33%).	14%	38%	33%
2018 Midterm Exit	D+8.6 wave year. Useful lower bound for midterm LV age compression.	13%	39%	33%

ACS/CPS Adult Pop.	All U.S. adults 18+. Heavily younger and non-college. Never used alone for LV targets.	~20%	~40%	~22%

Target differences between RV and LV universes: The LV targets will systematically differ from RV targets in predictable ways. Younger age groups (18–29) shrink by approximately 2–3pp in the LV universe because younger voters have lower and more volatile midterm turnout. The 65+ group expands to approximately 30% of the LV electorate — this cohort votes at the highest and most consistent rates. White College expands by approximately 2pp; Hispanic contracts by ~1.5pp due to historically lower midterm turnout. The DNV (Did Not Vote) recall cell compresses from ~19% in the RV universe to 6–8% in LV.

Phase 1 — Shared Data Foundation

This is the only phase shared by both the RV and LV tracks. After this phase completes, the pipeline diverges into two independent tracks that share no computations.

1

Data Ingestion & Quality Screening

Load the raw survey CSV from the field platform (Pollfish or equivalent). Run fuzzy column auto-detection: scan every column header for Q-prefix matches to identify question columns without relying on exact names. Apply a pre-weight quality pass before any demographic variables are derived. Flag speeders: respondents whose completion time falls below 30% of the median completion time. Flag straightliners: respondents who selected identical answers across battery questions (Q16, Q17). Remove all flagged rows. Derive all demographic variables needed by the raking engine: Age bucket (18–29 / 30–44 / 45–64 / 65+); Sex; AgeGender joint (8 cells); RaceEdu 5-cell (White split by college graduation, Black, Hispanic, Asian/Other); Education 4-way; GenderEdu joint; Region (8 Census regions via state FIPS mapping); Income (7 bands); Vote2024 bucket.

2

LV Score Computed on Raw Unweighted Sample

The Likely Voter propensity score is computed HERE — before any raking or weighting — on the unweighted raw sample. This is the critical architectural decision that separates Pathway 3 (and Pathway 2) from Pathway 1. If LV scoring happens after RV weighting, the LV probability is contaminated by RV weighting decisions. By computing it on unmanipulated data, the LV propensity reflects only the respondents' actual survey answers, not the weights assigned to them. Both the raw geometric LV score and the initial P(vote) logistic calibration are stored for each respondent.

3

Q2 Vote History Bucket Assignment

From Q2 (multi-select: list of elections the respondent voted in), assign each respondent to one of three history buckets: Consistent (voted in 3 or more of the listed elections); Occasional (voted in 1–2 elections); New (0 elections, or explicitly selected "I have never voted"). Store this bucket alongside the LV score. The history bucket modulates the logistic k parameter in the LV formula: consistent voters receive k=14 (steeper curve — their stated intent is more credible and predicts behavior), occasional voters receive k=12 (default), new voters receive k=8 (flatter curve — regress toward the mean more aggressively because aspirational intent is a weaker predictor for first-time voters). After this step the pipeline splits. The RV track and LV track share no further computations.

Deep Dive: The Three-Question Likely Voter Screen

The PSI LV screen uses three orthogonal questions — Q3, Q4, and Q5 — that together predict individual turnout more accurately than any single-question screen. Most commercial pollsters use one question. The three-question geometric design is one of the most methodologically sophisticated LV screens in use in public opinion research.

Why three questions, and why geometric mean? The three questions are designed to capture independent dimensions of turnout: motivational intensity (psychological commitment to vote), behavioral preparedness (whether intent has translated to concrete action), and social voting environment (peer reinforcement of voting norm). These dimensions are multiplicatively related — a very high score on motivation is not sufficient if behavioral preparedness is extremely low. The geometric mean enforces this multiplicative penalty: a respondent with Q3=1.0, Q4=0.289, Q5=1.0 receives a geometric score of 0.66, correctly depressed from the high motivation signal. An arithmetic mean would give ~0.76, masking the preparedness gap.

Q3 — Motivational Intensity

Q3 is the strongest single predictor of individual turnout in the academic literature. It captures the psychological commitment to vote — the internal state that converts intention into action. The response options are calibrated on a continuous [0,1] scale. The full-certainty response receives weight 1.000 while the certain-not-to-vote response receives 0.029 (near zero but nonzero, preserving the continuous model for all respondents). Note: The PSI platform uses a geometric mean formulation for Q3/Q4/Q5; the Wisconsin script uses an arithmetic blend (0.50×Q3 + 0.30×Q4 + 0.20×Q5) with Q2 history. The recommended synthesis modulates the logistic k by Q2 history bucket rather than blending.

Q3 Response Option	PSI Weight	Behavioral interpretation
I am certain to vote and highly motivated to do so	1.0000	Maximum credibility — psychological commitment fully present
I am very likely to vote and feel motivated	0.9000	High confidence — minor uncertainty only
I am somewhat likely to vote but not strongly motivated	0.6680	Moderate signal — motivation gap may suppress turnout
I am motivated but unsure if I will actually vote	0.3320	Mixed signal — intent present but barriers acknowledged
I am not very likely to vote and feel little motivation	0.1090	Weak signal — turnout unlikely; models near exclusion
I am certain not to vote	0.0290	Near-zero weight — preserved for mathematical continuity

Q4 — Behavioral Preparedness

Q4 captures whether voting intent has translated into concrete action. A respondent who is highly motivated but has not yet determined how or where to vote is a significantly weaker turnout predictor than one who has confirmed their polling location or received their mail ballot. Research consistently shows that intention without preparation is a poor predictor of actual behavior. Q4 directly operationalizes this finding by differentiating between resolved and unresolved voting logistics.

Q4 Response Option	PSI Weight	Preparedness signal
In person on Election Day — I know my polling location	1.0000	Fully prepared — logistics resolved
Early in-person voting — I know when and where early voting is available	0.9370	Fully prepared with scheduling certainty
Mail-in or absentee ballot — I have already requested or received my ballot	0.9730	Highest certainty — ballot physically in hand or filed
In person on Election Day — I still need to confirm my polling location	0.7110	Intent firm; minor logistical gap remains
Early in-person voting — I still need to look up early voting details	0.5000	Intent present; meaningful logistical uncertainty
Mail-in or absentee ballot — I plan to request one but haven't yet	0.5000	Intent present; action step not yet completed
I haven't decided how I will vote yet	0.2890	Unresolved logistics — meaningful turnout risk
I do not plan to vote	0.0630	Near-exclusion; preserves continuity of model

Q5 — Social Voting Environment

Q5 captures the social reinforcement of voting intention. Peer voting behavior is one of the strongest predictors of individual turnout at the individual level (Gerber & Green, 2000; Nickerson, 2008). A respondent embedded in a social network where voting is normative is substantially more likely to follow through on stated intent than one in a non-voting social environment. This dimension is orthogonal to both Q3 and Q4 — it represents an external constraint rather than an internal disposition.

Q5 Response — "Of the 5–10 people closest to you..."	PSI Weight	Social norm signal
All or nearly all of them plan to vote	0.8910	Strongest social norm — universal peer reinforcement
Most of them plan to vote	0.9500	Strong positive social norm
About half of them plan to vote	0.5000	Neutral social environment; no directional norm
A few of them plan to vote	0.0150	Weak/negative social signal — non-voting norm present
Not sure	0.2690	Ambiguous signal — treated conservatively

None of them plan to vote	0.1190	Negative social norm — active disincentive to vote

The PSI Geometric Mean Formula

Raw LV score = cube_root(Q3_weight × Q4_weight × Q5_weight) This is implemented as: LV_raw = (w3 × w4 × w5)^(1/3) The geometric mean is mathematically correct here because it penalizes any single low score multiplicatively. If a respondent scores 1.0 on motivation and social environment but only 0.063 on preparedness (does not plan to vote), the geometric result is (1.0 × 0.063 × 1.0)^(1/3) = 0.398 — substantially suppressed. The arithmetic mean would give 0.688 — dangerously overstating turnout probability.

Logistic Calibration: Converting Raw Scores to P(vote)

The raw geometric LV score is not a probability of voting — it is a relative propensity on [0,1]. To convert it to an actionable probability, the PSI system passes it through a logistic S-curve calibrated so that the weighted area under the curve equals the projected turnout rate for the cycle.

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P(vote) = 1 / (1 + e^(-k × (LV_raw - μ)))
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Parameter	Value	How it is determined
μ (midpoint)	Solved numerically	60-iteration binary search so that Σ P(vote)×weight_rv / Σ weight_rv = projected turnout rate
k (steepness)	14 / 12 / 8	Varies by Q2 history bucket: Consistent k=14, Occasional k=12, New k=8. Steeper k = sharper separation between certain and uncertain voters for more credible respondents
Target turnout rate	117M / 173.8M = 67.3%	2026 projected: 117 million voters out of 173.8 million registered. Adjustable in application UI

Why k varies by history bucket: A consistent voter's stated intent in Q3 is highly credible — past behavior is the strongest predictor of future behavior. A first-time voter who says "certain to vote" is aspirational: their stated intent has no track record behind it, and should be regressed toward the mean more aggressively. Setting k=14 for consistent voters and k=8 for new voters accomplishes this without changing the response weight maps — it changes how sharply the model distinguishes between P=0.7 and P=0.9 for each group.

Q2 Vote History — The Recommended Synthesis

Q2 asks respondents to select which elections they have voted in from a list spanning 2014–2024 (up to 6 elections). The Wisconsin script uses Q2 as a 40% blend component in an arithmetic LV formula. The PSI platform uses only Q3/Q4/Q5 and ignores Q2. The recommended Pathway 3 synthesis uses Q2 to modulate the logistic k parameter — keeping the continuous geometric model while using vote history to validate the credibility of stated intent.

Q2 bucket	Qualifying elections	Logistic k	Effect
Consistent	3 or more elections	k = 14	Steepest curve — stated intent most credible. Sharp separation between 0.9 and 0.7 P(vote).
Occasional	1–2 elections	k = 12	Default curve — moderate regression to mean. Balances stated intent with track record.
New voter	0 or "never voted"	k = 8	Flattest curve — strongest regression to mean. Aspirational intent heavily discounted.

Phase 2 — Post-Field SOCAL Target Derivation

Once the field closes, the pre-survey prior targets established in Phase 0 are updated using the SOCAL (Survey-Observed Calibrated Adaptive Likelihood) credibility-weighted procedure. This update is applied separately and independently for the RV universe and the LV universe.

2a

RV Target Formation — SOCAL Credibility Update

For each demographic cell, compare the pre-survey prior proportion to the observed (unweighted) sample proportion. If the difference is 3 percentage points or less, the final target remains at the prior value. The survey observation is treated as noise within the margin of prior uncertainty. If the difference exceeds 3pp, apply the credibility-weighted update: final_target = (0.70 × prior) + (0.30 × observed). This formula places 70% credibility weight on the external benchmark and 30% on what the survey actually captured. It prevents extreme raking adjustments that would arise if the prior were held rigidly against a substantially different sample, while respecting the external benchmark more than the survey. After this step, all RV targets are locked. No further target adjustment is permitted.

2b

LV Target Formation — P(vote)-Weighted SOCAL Update

The LV target formation uses the same 70/30 SOCAL credibility formula, but the prior and the observed composition are both defined differently. The prior for the LV universe is formed from two prior midterm exit polls (2022 at 50%, 2018 at 50%) rather than the ACS/exit blend. The "observed" for LV is not the raw sample composition — it is the P(vote)-weighted composition. For each demographic cell, multiply each respondent's cell membership indicator by their computed P(vote), sum across cells, and normalize. This produces LV targets that reflect which demographic groups in this specific sample are expected to turn out in the 2026 cycle. The SOCAL formula then blends this data-driven estimate with the midterm electorate prior.

Final Weighting Set A — RV Target Specification (2026 Midterm)

Dimension	Type	Cells	RV Targets	LV Adjustment
Age × Sex	Joint	8	18–29: M 7.9% / F 8.1% · 30–44: M 12.3% / F 13.3% 45–64: M 15.1% / F 16.2% · 65+: M 12.8% / F 14.3%	18–29 compressed ~2.5pp (M+F). Redistributed to 65+. LV 65+ rises to ~30%.
Education × Sex	Joint	4	Male No College 26.5% · Male College 21.7% Female No College 24.9% · Female College 26.9%	~2pp shift No College→College per sex. LV Female College rises to ~28–29%.
Race × Education	Joint (5-cell)	5	White No College 34.5% · White College 32.5% Black 11.0% · Hispanic 11.5% · Asian/Other 10.5%	White College +2pp. Hispanic -1.5pp (lower midterm turnout historically).

Region	Marginal	8	NE 4.8% Mid-Atl 15.3% SE Atl 15.8% Appalachia 7.7% Gt Lakes 16.7% LMW/Plains 5.7% SW 12.2% West 21.8%	Modest shifts: West/NE expand slightly, SW contracts (turnout differentials).
2024 Recall	Marginal	4	Trump 40.35% · Harris 39.69% · Third 1.44% · DNV 18.52%	LV: DNV compressed to 6–8%. Voter shares rescaled proportionally. No CPS anchor for LV.

Phase 3a — RV Track: Wisconsin Pipeline

The RV track runs the complete Wisconsin statistical pipeline against the locked RV targets. It has no knowledge of LV scores, LV targets, or any LV computation. The output — weight_rv — is the definitive registered voter electorate weight.

3a-1

Cell Collapse Safeguard

Before entropy balancing or raking begins, check all raking cells against minimum n thresholds. Joint cells (AgeGender, GenderEdu) require $n \geq 20$ unweighted respondents. Marginal cells require $n \geq 15$. Thin cells are collapsed into the nearest neighbor by target proportion similarity — they are never dropped. When a joint cell collapses, the target proportions of both cells are merged into the surviving cell. All collapse decisions are logged for the audit trail. This prevents the raking algorithm from producing extreme weights trying to satisfy an empty cell constraint.

3a-2

Entropy Balancing Initialization (Hainmueller 2012)

Run entropy balancing: minimize KL-divergence from uniform weights subject to moment constraints from the RV targets. This produces a starting weight vector that satisfies all target moments exactly before raking begins — a much stronger initialization than uniform weights. Optimizer: L-BFGS-B. Fallback: uniform initialization if optimizer fails to converge. Cap raw EB weights at 3.5× the mean before passing to raking. This cap is critical: entropy balancing can produce extreme weights on thin cells, and those extreme starting weights would destabilize raking convergence.

3a-3

Iterative Raking — 4 Rounds with DEFF-Informed Cap

Run IPF raking against all active dimensions from the chosen weighting set. Set A: AgeGender, GenderEdu, RaceEdu, Region, Recall2024 (33 cells total). Set B adds Age×Education joint dimension (41 cells). Round 1: rake with no weight cap. After round 1, compute natural DEFF. Set cap: if DEFF < 2.5, cap weights at 1.75× the mean; else cap at 2.0×. Rounds 2–4: after each pass, apply cap, renormalize, continue. Stop early if zero weights exceed cap. Convergence threshold: max cell deviation < 0.01% across all raking dimensions. Convergence diagnostics printed after every round.

3a-4

Two-Stage 2024 Recall Calibration

This step treats voters and non-voters separately. Conflating them creates two structurally different recall error types. Stage 1 — Voters: among respondents who reported voting in 2024, rake to FEC certified popular vote shares: Trump 49.91% / Harris 48.39% / Third 1.70%. Cap per-respondent adjustment at 2.0× to prevent single-respondent influence. Stage 2 — Non-voters: adjust the total Did Not Vote share to the CPS 2024 ASEC estimate (~35% non-voter share). Trim post-calibration weights at the 99th percentile. Renormalize to original weight sum after each stage.

3a-5

Response Propensity Scoring

Fit a logistic regression predicting panel participation against a synthetic population drawn from the demographic targets. Covariates: AgeGender, Gender, Education4, RaceEdu. Flag respondents with propensity score < 0.10 as extreme non-responders. Propensity scoring is a diagnostic output: report count and percentage of extreme non-responders. If more than 5% of the sample falls below 0.10 propensity, apply inverse propensity trimming as a pre-weight before the raking pass. Final output: weight_rv.

Phase 3b — LV Track: Independent Wisconsin + PSI Pipeline

The LV track is architecturally independent from the RV track. The starting weights for entropy balancing are the P(vote) scores computed in Phase 1 — not uniform weights and not weight_rv. This means turnout propensity shapes the LV universe from the very first optimization step. No weight_rv value enters the LV track at any point.

3b-1

Final LV Propensity — Synthesis Formula

Compute the final calibrated LV probability for each respondent using the recommended synthesis: Step 1: LV_raw = cube_root(Q3_weight × Q4_weight × Q5_weight) using PSI response weight maps. Step 2: Assign history bucket from Q2 → k value: Consistent k=14, Occasional k=12, New k=8. Step 3: Solve μ numerically per history bucket via 60-iteration binary search so that the weighted turnout rate equals the projected share for that bucket. Step 4: P(vote) = 1 / (1 + e^(-k × (LV_raw – μ))). Every respondent retains a non-zero P(vote) and contributes to the LV universe proportionally. There is no binary inclusion/exclusion threshold — the continuous probability preserves all information.

3b-2

LV Weight Initialization — P(vote) as Starting Weight

The initial weight for the LV track is: weight_lv_init = P(vote). This is the key architectural departure from the RV track. In the RV track, entropy balancing starts from uniform weights (every respondent equally likely). In the LV track, it starts from the P(vote) distribution — respondents who are more likely to vote carry more influence from the very first step. Normalize so that mean(P(vote)) = 1.0 before entropy balancing begins.

3b-3

LV Entropy Balancing — P(vote) Seeded

Run Hainmueller entropy balancing against the LV-specific targets from Phase 2b. Starting weights are the normalized P(vote) values from step 3b-2 — not uniform. The optimizer minimizes KL-divergence from the P(vote)-shaped distribution rather than from uniform. This means the EB result respects both the turnout propensity structure AND the demographic targets simultaneously. Apply the same 3.5× raw EB weight cap before passing to raking.

3b-4

LV 4-Round Raking Against LV-Specific Targets

Run the same 4-round raking pipeline as the RV track, but against the LV-specific demographic targets from Phase 2b. Starting weights are the LV EB-initialized weights (P(vote) seeded). This is weighted raking (rakingWeighted in the PSI platform) — it begins from a non-uniform initialization and adjusts proportionally. Convergence and cap logic are identical to the RV pass. The result is an LV weight that is independently balanced for the expected LV electorate.

3b-5

LV Recall Calibration — Voter Shares Only (No CPS DNV Anchor)

Recall calibration for the LV track uses FEC certified voter shares only: Trump 49.91% / Harris 48.39% / Third 1.70%. There is no CPS DNV anchor for the LV track. The rationale: the LV screen through P(vote) scoring has already suppressed likely non-voters through the propensity mechanism. Adding a CPS DNV correction would double-count that suppression — it would artificially compress the Did Not Vote cell below what is meaningful for the LV electorate. Trim post-calibration weights at the 99th percentile. Renormalize. Final output: weight_lv — completely independent of weight_rv.

Phase 4 — Monte Carlo Sensitivity Analysis

The Monte Carlo analysis is unique to Pathway 3. It quantifies the honest uncertainty envelope for LV estimates by running the full LV pipeline across 9 scenarios. These results must be disclosed in methodology notes whenever LV results are published.

> The 9-scenario grid crosses three projected turnout levels against three LV demographic target sets. For each combination, the complete LV raking pipeline reruns from the EB initialization step. Key toplines (generic ballot, Trump approval, right track) are recorded for each scenario. The spread across all 9 scenarios constitutes the honest uncertainty envelope. For 2026 generic ballot polling this range is typically ±2–3 percentage points.

	Low turnout: 105M	Base turnout: 117M	High turnout: 128M
2022-anchored targets (prior midterm exits)	Scenario 1	Scenario 2 ← Base	Scenario 3
P(vote)-derived targets (data-driven from sample)	Scenario 4	Scenario 5	Scenario 6
PSI composite targets (2022 + 2018 blend)	Scenario 7	Scenario 8	Scenario 9

The PSI platform implements Monte Carlo via probabilistic inclusion: for each simulation, each respondent is included with probability proportional to their LV score blended with a certainty component. The included subset is raked against the base targets. Results from 5,000 simulations yield percentile-based confidence intervals (default 90%).

Phase 5 — Merged Diagnostics & Dual-Universe Output

All diagnostics are run independently for both universes. No result from one universe explains or adjusts the other.

Output	Universe	Purpose and interpretation
DEFF & Kish DEFF	Both (separate)	Design effect measuring weight variance. DEFF < 1.5 = good; > 2.0 = review before publication.
Effective N	Both (separate)	n / DEFF — the real statistical power of the weighted sample. Used for margin of error.
Covariate balance (SMD)	Both (separate)	Standardized Mean Difference < 0.10 = balanced for each demographic dimension.
Bootstrap SE (500 resamples)	Both (separate)	Empirical standard errors for key questions: generic ballot, Trump approval, right track.

ICC / geographic clustering	Both (separate)	Intraclass correlation at the regional level. DEFF_cluster = 1 + (m■-1) × ICC.
RV/LV shift decomposition	Comparison	Three-column table: (1) RV composition, (2) post-P(vote) before re-raking, (3) final LV. Decomposes screen effect vs. weighting artifact.
Topline tabbooks	Both (separate)	Complete weighted frequency tables for all questions in RV and LV universes.
Monte Carlo CI	LV only	9-scenario range alongside point estimates. Must be disclosed in methodology notes.
External validation	Both (separate)	Trump approval vs. RCP aggregate. If deviation > 3pp, investigate before publication.

When to Use Pathway 3 vs. Pathways 1 and 2

Attribute	Pathway 1	Pathway 2	Pathway 3
LV computed when	After RV weighting	Before RV weighting	Before anything
LV demographic correction	None — inherits RV	Yes — 2nd rake pass	Yes — independent rake
EB coverage	RV only	RV only	Both independently
Monte Carlo	Turnout only	Turnout only	Full 9-scenario grid
Implementation effort	Low	Medium	High
Min recommended n	600+	800+	1,000+
Best weighting set	A or C	A or B	B preferred
Publication defensibility	Strong	Very strong	Highest
Ideal use case	Production default, benchmark wave	Publishing both RV+LV, competitive race	Peer review, methodology paper, high-scrutiny publication

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PART II

Application Architecture Specification

This section describes the modules and user flows of the polling analysis application that implements the Pathway 3 methodology end-to-end. The application accepts a raw survey CSV or Excel file, guides the user through every methodology step described in Part I, and produces interactive displays and exportable professional outputs.

MODULE OVERVIEW

Module	Function	Methodology phase
1. Data Upload & Raw Explorer	Ingest CSV/Excel, quality screen, display unweighted data with bar charts, pies, frequency tables	Phase 1 — Ingestion
2. LV Model Panel	Display Q3/Q4/Q5 response weight maps; compute and visualize LV score distribution; logistic calibration controls	Phase 1 — LV scoring
3. Benchmark Setup	Pre-survey target entry or preset selection; SOCAL prior-posterior update; Q2 history parameters	Phases 0 & 2 — Target formation

4. Weighting Engine	Run full Pathway 3 pipeline: EB, 4-round raking, recall calibration, both tracks	Phases 3a & 3b
5. RV Universe Display	Weighted toplines, bar charts, diagnostics for Registered Voter universe	Phase 5 — RV output
6. LV Universe Display	Weighted toplines with Monte Carlo CI bands, diagnostics for Likely Voter universe	Phases 4 & 5 — LV output
7. Cross-Tab Builder	Build, display, and export any cross-tabulation in either universe; save to named library	Phase 5 — Extended tabbook
8. PDF Report Generator	Export topline and/or cross-tab results as formatted presentation PDF	Publication output

Module 1 — Data Upload & Raw Data Explorer

The user uploads a raw CSV or Excel file from the survey platform. The module auto-detects question columns using fuzzy Q-prefix matching and displays the raw, unweighted data.

- Auto-detection of Q3, Q4, Q5, Q2 history columns, demographic columns using fuzzy header scan.
- Quality screening report displayed immediately: speeders flagged, straightliners flagged, rows removed.
- User confirms or adjusts quality screening thresholds before proceeding.
- Bar chart per question: response option on x-axis, unweighted count/percentage on y-axis.
- Pie chart for categorical demographics (Sex, AgeGroup, Region, RaceEdu).
- Sortable frequency table for every question with raw counts and unweighted percentages.
- Demographic composition panel: observed sample vs. selected benchmark targets (flags imbalances).

Module 2 — LV Model Panel

This module exposes the full PSI LV model to the user. It is the direct implementation of the three-question screen described in Phase 1.

- Displays the Q3, Q4, and Q5 response weight maps as editable tables — users can see and modify each response's weight.
- Respondent count shown per response category so users understand how many respondents each weight applies to.
- LV score distribution chart: histogram of raw geometric scores and calibrated P(vote) probabilities shown side-by-side.
- Logistic calibration controls: total registered voters, expected turnout, logistic k parameter.
- Real-time update: any change to weights or parameters immediately recomputes LV scores and redraws the distribution.
- Displays calibrated μ (logistic midpoint), AUC (model-implied turnout rate), mean P(vote), counts of respondents with $P \geq 0.9$ and $P \leq 0.1$.

- Q2 history bucket assignment displayed: breakdown of sample by Consistent / Occasional / New voter.

Module 3 — Benchmark Setup & Target Derivation

Before weighting runs, the user establishes the demographic benchmark targets for both universes. This implements Phases 0 and 2.

Input	Required from user	Methodology step
Election cycle	Midterm or presidential, year. Selects correct exit poll for SOCAL prior.	Phase 0
Weighting set	Set A (33 cells), Set B (41 cells), or Set C (21 cells). B recommended for Pathway 3.	Phase 3a/3b
RV benchmark preset	PSI composite (default), 2022 exit, 2024 exit, ACS/CPS, or custom entry.	Phase 0 / 2a
LV benchmark approach	P(vote)-derived auto-generation, PSI composite, 2022+2018 blend, or custom entry.	Phase 0 / 2b
Projected turnout	Base turnout (default 117M / 173.8M = 67.3%). Low/high for Monte Carlo.	Phase 3b-1
k parameter	Consistent k=14 / Occasional k=12 / New k=8. User-adjustable for sensitivity.	Phase 3b-1
FEC recall targets	Pre-loaded: Trump 49.91% / Harris 48.39% / Third 1.70%. User-adjustable.	Phase 3a-4 / 3b-5
CPS non-voter anchor	CPS 2024 ASEC DNV share for RV recall Stage 2. Not used in LV track.	Phase 3a-4
Column mapping	Verify auto-detected Q3/Q4/Q5/Q2 column mappings. Confirm or reassign.	Phase 1

Module 4 — Weighting Engine

The engine runs the full Pathway 3 pipeline after the user confirms all inputs. Progress is shown step by step. Convergence logs are displayed for audit.

- Phase 1: Quality screening applied. LV scores computed on raw sample. Q2 history buckets assigned.
- Phase 2a/2b: SOCAL credibility update applied independently for RV and LV targets. Final targets locked.
- Phase 3a: RV track — cell collapse → EB initialization → 4-round raking → two-stage recall → propensity → weight_rv.
- Phase 3b: LV track — LV propensity synthesis → P(vote)-seeded EB → 4-round LV raking → voter-only recall → weight_lv.
- Phase 4: Monte Carlo 9-scenario grid. Topline variance stored for key questions.

- Phase 5: DEFF, effective N, covariate balance, bootstrap SE, shift decomposition, external validation computed.
- Warning displayed if DEFF > 2.0 for either universe — flagged for review before publication.
- Convergence log available for download as part of methodology disclosure package.

Modules 5 & 6 — RV and LV Universe Displays

Two parallel displays — identical structure, independent data. The LV display adds Monte Carlo CI.

- Horizontal bar chart per question: weighted % per response, with unweighted comparison toggle.
- Response table: weighted %, unweighted %, weighted n, effective n, bootstrap SE.
- LV display: Monte Carlo range shown as shaded error band on bar charts, with low/high scenario annotation.
- Diagnostics panel: DEFF, Kish DEFF, effective N, weight distribution histogram, covariate balance SMD.
- RV/LV shift summary (comparison view): three-column decomposition of screen effect vs. weighting artifact.
- External validation: side-by-side of survey result vs. RCP aggregate for Trump approval and generic ballot.

Module 7 — Cross-Tab Builder & CSV Exporter

- Select banner variable (column header): any demographic dimension or survey question.
- Select stub variable (row): any survey question.
- Select universe: RV weighted, LV weighted, or both side-by-side.
- Statistic type: column % (default), row %, or weighted n.
- Optional subgroup filters: e.g., only respondents in a specific region or age group.
- Net Difference column: RV minus LV per cell (when both universes shown).
- Statistical significance flagging: cells that differ significantly from the column total highlighted.
- Effective n per banner column displayed to identify thin subgroups (< 30 flagged).
- Export current cross-tab as CSV with full metadata header (survey name, date, universe, weight parameters).
- Export all saved cross-tabs as single multi-sheet CSV package.
- Export full weighted dataset as respondent-level CSV with weight_rv and weight_lv columns appended.

Module 8 — PDF Report Generator

Report type	Contents	Use case
Topline Report	Cover page · executive summary · all topline questions with bar charts and weighted tables · RV/LV shift summary · methodology disclosure · diagnostics appendix · Monte Carlo CI disclosure	Client delivery, press release, publication submission
Topline + Cross-Tab Report	All of the above, plus all saved cross-tabulations formatted with full demographic breakdowns in both universes	Internal research, peer review, detailed client deliverable

- Survey name, client name, date, and uploaded logo populate the cover page automatically.
- User selects which questions to include in the topline section.
- Universe selection: RV only, LV only, or both with shift comparison columns.
- Methodology disclosure page auto-populated from the pipeline run parameters (targets, k values, turnout, weighting set).
- Monte Carlo CI disclosure auto-included whenever LV results are present.
- External validation note included when either universe deviates > 3pp from RCP aggregate.

Pathway 3 in one sentence: Two fully independent pipelines — each initialized with its own pre-survey targets, its own SOCAL credibility update, its own entropy balancing pass, its own four-round raking against universe-specific demographic targets, and its own recall calibration — converge only at the final reporting stage, where they are presented side-by-side with a Monte Carlo uncertainty envelope, a three-way shift decomposition, and a complete diagnostic disclosure. This represents the highest standard of methodological rigor available in commercial public opinion research.